Pure Shooting %: Fixing Basketball's Most-Used Efficiency Stat

Four box score numbers. No approximations. A true 0-100% scale.

The Problem

Shooting efficiency measures one thing: of all the points a player could have scored, how many did they actually score? Basketball has three ways to score — 2-point field goals, 3-point field goals, and free throws — and True Shooting Percentage (TS%) is the NBA's standard attempt to combine all three into a single number. But the formula has a structural flaw. Its denominator treats every field goal attempt as worth a maximum of 2 points. A 3-pointer is worth 3. This means a player who makes every single 3-point attempt gets a TS% of 150% — not 100%. A player who makes every 2-point attempt gets 100%. Both shot perfectly.

Scenario	PTS	FGA	FTA	TS%	PS%
10/10 on 2-pointers	20	10	0	100.0%	**100.0%**
10/10 on 3-pointers	30	10	0	150.0%	**100.0%**
10/10 on free throws	10	0	10	113.6%	**100.0%**

All three players scored every possible point. Pure Shooting % (PS%) gives all three 100%. TS% ranges from 100% to 150% depending on shot type.

The Fix

PS% = PTS / (2 x FGA + 3PA + FTA)

Points scored divided by maximum points possible. The denominator accounts for the actual maximum value of each attempt: 2 per 2PT, 3 per 3PT, 1 per FT. The full derivation uses only the identity 2PA = FGA - 3PA.

47 Seasons of Distortion

Season	League 3PA Rate	PS%	TS%	Delta
2025-26	41.6%	**48.3%**	58.0%	-9.7pp
2024-25	41.8%	**48.1%**	57.8%	-9.7pp
2015-16	28.5%	**49.2%**	55.9%	-6.7pp
2005-06	21.4%	**49.3%**	54.4%	-5.1pp
1996-97	22.2%	**48.2%**	53.5%	-5.3pp
1989-90	10.4%	**49.2%**	51.8%	-2.6pp
1979-80	3.1%	**49.6%**	51.1%	-1.5pp

The distortion grew from -1.5pp to -9.7pp as the league 3PA rate rose from 3% to 42%. 3PA rate explains 99.86% of the variance (R^2 = 0.9986).

Distortion Across Eras

Player	Season	3PA Rate	TS%	PS%	Distortion
Kareem Abdul-Jabbar	1979-80	0.1%	63.9%	**62.7%**	+1.1pp
Michael Jordan	1990-91	5.1%	60.5%	**58.1%**	+2.4pp
Shaquille O'Neal	1999-00	0.1%	57.8%	**56.4%**	+1.4pp
Ray Allen	2001-02	46.0%	59.8%	**49.0%**	+10.9pp
Kobe Bryant	2005-06	23.8%	55.9%	**49.8%**	+6.1pp
LeBron James	2012-13	18.8%	64.0%	**58.2%**	+5.8pp
Steph Curry	2015-16	55.4%	66.9%	**53.0%**	+13.9pp
James Harden	2018-19	53.9%	61.6%	**49.4%**	+12.2pp
Nikola Jokic	2023-24	16.3%	65.1%	**59.7%**	+5.3pp
Shai Gilgeous-Alexander	2025-26	22.5%	66.8%	**59.8%**	+7.0pp

Rim scorers (Kareem, Shaq) are barely affected. High-volume 3-point shooters (Curry, Harden) are overrated by 10+ percentage points. The distortion is a direct function of 3PA rate.

At the current league 3PA rate of ~42%, the average distortion is 9.7 percentage points. If the league reaches 50%, it will exceed 11. The gap is not stabilizing. It is growing.

TS% is not bounded at 100%, cannot be compared across archetypes or eras, and is not a true percentage. PS% eliminates these problems with a simpler formula and a true 0-100% scale.

PS% data available for all NBA players, 47 seasons (1979-80 through 2025-26)